



Freeze-drying of Active Pharmaceutical Ingredients

Employing the BUCHI Lyovapor™ L-250 for efficient freeze-drying

The synthesis and purification of related substances (RS) of Active Pharmaceutical Ingredients (APIs) are vital steps in drug development, ensuring the production of therapeutic compounds with high purity and yield. This process involves optimizing reaction conditions and employing techniques such as liquid-liquid extraction, filtration, crystallization, chromatography, distillation, and freeze-drying to isolate and purify the API-RS. Given the temperature sensitivity of the samples, freeze-drying is used as a safe and effective method to remove moisture, enabling longer preservation. These techniques are crucial for maintaining the structural integrity of sensitive compounds and ensuring compliance with regulatory standards.

Keywords: Freeze-drying, APIs, BÜCHI Lyovapor™ L-250.

Introduction

The synthesis of small molecules (molecular weight less than 1000 Da or g/mol), such as Active Pharmaceutical Ingredients (APIs) and their related substances (RS), is a pivotal step in drug development. This process involves producing the core chemical compound responsible for the therapeutic effect. It requires a combination of organic chemistry, chemical engineering, and analytical techniques to ensure that the final product meets stringent quality standards.

A crucial aspect of API-RS synthesis is optimizing reaction conditions to achieve consistent and reproducible results. Once the reactions are completed, a first purification isolates the API from by-products, impurities, and unreacted materials.

The workup process typically begins with quenching, where the reaction is stopped by neutralizing reactive intermediates, often using water, an acid, or a base. This

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stabilizes the product and removes inorganic by-products, commonly through liquid-liquid extraction. Following extraction, filtration is employed to remove solid impurities or residual catalysts. Depending on the material and particle size, techniques such as gravity filtration, vacuum filtration, or pressure filtration may be used.

Subsequent purification techniques, including crystallization, distillation, and chromatography, are employed to achieve the desired purity. Crystallization isolates solid APIs by dissolving the crude product in a suitable solvent and inducing crystal formation under controlled conditions. For more complex mixtures, chromatography is used, providing precise separation of the target compound from closely related impurities. Purification is essential for meeting regulatory requirements concerning potency, purity, and stability, as well as ensuring the therapeutic efficacy and safety of the final drug product.

After purifying temperature-sensitive materials using normal-phase or reverse-phase chromatography, it is essential to remove any residual solvents. Freeze-drying is preferred over other drying technologies for this purpose, as it effectively removes solvents while preserving the integrity of the final purified material. Freeze-drying is significant for stabilizing thermally/heat-sensitive products or those that could degrade during traditional drying methods including air drying, vacuum drying or the use of desiccators. The resulting final sample is preserved for further analysis or studies.

Equipment, Chemicals and Materials

- › PrepChrom™ C-700, BÜCHI Labortechnik AG.
- › C18 column (10 µm).
- › Lyovapor™ L-250 Basic, BÜCHI Labortechnik AG.
- › Advanced vacuum control.
- › Manifold drying chamber with 12 ports.
- › Two-stage Rotary vacuum pump.
- › Rotary evaporator with Dewar container.
- › Acetone.

- › Dry ice.
- › Liquid Nitrogen.
- › Round Bottom flask (50, 100, 200, 500 & 1000mL).
- › Acetonitrile 99.9%, Honeywell.
- › Ethanol 90%, Merck.
- › Methanol 90%, Merck.
- › Trifluoro acetic acid 99.9%, Carl Roth.
- › Acetone 99.5%, Honeywell.
- › Acetic acid 95.9%, Carl Roth.

Purification & Isolation

Preparative Liquid Chromatography (Prep LC) is designed to isolate compounds in quantities and with purity sufficient for further analysis. This process is essential in various fields and aims to achieve a final purity of $\geq 95\%$ for the compound of interest.

A. Modes of Separation

Separation was achieved using reversed-phase chromatography, the most commonly employed technique in liquid chromatography. In this method, the stationary phase is less polar than the eluting solvent, which typically consists of a mixture of water and organic solvents, such as acetonitrile or methanol.

B. Quick Start Method Development

For reversed-phase separations, method development typically starts with a C18 column (10 µm, 30 x 300 mm) to optimize the solvent system for the target compound, considering the use of acidic or basic pH buffers. The mobile phase is prepared in two components: an aqueous phase (designated as Mobile Phase A) and an organic solvent (designated as Mobile Phase B). This phase consists of a mixture of solvents, including acetonitrile (60–80%), methanol (10–60%), trifluoro acetic acid (TFA), acetic acid and water.

C. Fraction Collection

Fraction collection can be performed manually or automatically, with automation often triggered by a UV detector signal. One common automated approach is the peak-based collection, where fractions are collected

D. Fraction Analysis and Recovery

After collection, fractions are evaluated for purity using HPLC and recovery methods to assess the efficiency and yield of the purification process. Recovery methods, such as UV spectroscopy, mass spectrometry (MS), and nuclear magnetic resonance (NMR), focus on quantifying the amount and verifying the purity of the target compound in the collected fractions. Recovery calculations are performed to determine the yield.

The recovery of a compound is typically calculated as a percentage of the initial amount of the compound applied to the system.

Recovery rate (%) =

$$\frac{\text{Amount of target compound in collected fractions}}{\text{Initial amount of target compound loaded onto the column}} \times 100$$

Freeze-drying

The start-up of Lyovapor™ L-250 instrument begins with the conditioning, including the cooling of the ice condenser and the vacuum pump warm-up. This step requires approximately 30 minutes.

Meanwhile the samples were prepared for freezing. The freezing procedure varies depending on solvent mixture. The samples containing acetonitrile as the main solvent were frozen in a mixture of acetone and dry ice at around -75 °C. For samples containing ethanol or methanol, a rotary evaporator was used to remove these solvents. This step was essential because methanol's low freezing temperature and potential evaporation under reduced pressure could complicate the freeze-drying process. The remaining solvents were frozen using liquid nitrogen at -196 °C.

All sample in the round-bottom flask was frozen using Dewar accessory and a rotary evaporator.

This rotational freezing technique created a uniform, thin layer with a larger surface area, significantly

reducing processing time and facilitating faster sample drying. The drying process was performed at a set vacuum pressure of 0.50 mbar. The samples were loaded onto the manifold valve singularly. After the vacuum reached the set value, or within 15 minutes, the next sample was attached to the manifold. This procedure was repeated for each sample, ensuring low actual pressure in the drying chamber that the samples are exposed to.

Results and Discussion

The following molecules are freeze-dried to remove the solvents and solvent residuals from the sample. The obtaining API-RS can be used in disease treatment as stated in Table 1.

The duration of the process ranged from approximately 4 to 20 hours, depending on the sample volume and the solvent composition. Generally, water requires a longer drying time compared to organic mixtures. No samples showed signs of collapse or melting during the process. The goal of this study to freeze-dry various samples were successfully processed using the Lyovapor™ L-250 Basic.



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L-250



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Purpose	API - RS Sample	Removing Solvent
Bacterial Infections	Amoxicillin	ACN, water
	Cefixime	ACN, water
	Cetirizine lactose ester	ACN, water
	Cephalexin	Water
	Ciprofloxacin	ACN, water
	Clindamycin	ACN, water
	Doxycycline	ACN, water
	Ertapenem	ACN, water, acetic acid
	Gentamicin	ACN, water
	Linezolid	ACN, water
	Minocycline	ACN, buffer, water
	Moxifloxacin	ACN, water
	Tetracycline	ACN, water
	Tigecycline	ACN, water
High blood pressure	Oxytetracycline	ACN
	Streptomycin	ACN, water
	Acebutolol	ACN, water
	Atenolol	Water
	Bisoprolol	Water
	Diltiazem	ACN
Diabetes	Labetalol	ACN, water
	Carvedilol	ACN, water
	Glimepiride	ACN, water
	Linagliptin	ACN, water
	Metformin	Water
	Sitagliptin	Water
Depression	Varicose glucose	ACN, water
	Voglibose	Water
	Duloxetine	Water
	Escitalopram	Water
	Fluvoxamine	Water
Cancer treatment	Vortioxetine	Water
	Vilazodone	ACN, water
	Cyclophosphamide	Water
Bladder treatment	Hydroxyethyl pyrrolidine	Water
	Hydroxyurea	Water
	Pazopanib	ACN, water
Hypocholesterolemia	Mirabegron	ACN, water
	Solifenacin	ACN, water
Treatment influenza A and B	Atorvastatin	ACN, water
	Bempedoic acid	ACN, water
	Laninamivir	ACN, water
Stomach ulcers	Oseltamivir	Water
Mouth ulcers	Famotidine	ACN, water
Anesthetic	Apremilast	ACN, water
Calcium deficiency	Bupivacaine	ACN, water
Malarial infections	Calcium Gluconate	ACN, water
Blood clots	Chloroquine	Water
Eye disease	Dabigatran	ACN, water
Insomnia	Dorzolamide	ACN, water
Prevent blood from clotting	Doxylamine	ACN, water
Fungal infection	Ethylenediamine tetraacetic acid (EDTA)	ACN, water
Anaemia deficiency	Efinaconazole	ACN, water
Treatment for leishmaniasis	Folic acid	Water
HIV viral particles	Meglumine	ACN, water
Delay spray	Mercapto benzamide	Water
Improving muscle strength	Dapoxetine	ACN, water
Eye drops	Neostigmine	ACN, water
Pain relief in newborns.	Nepafenac	Water
Analgesics	Aminophenol	ACN, water
Relief of stuffy nose, sinus and ear symptoms	Paracetamol	ACN, water
Mood disorders	Phenylephrine	Water
Migraine headaches	Risperidone	Water
To lower phenylalanine levels in the blood	Rizatriptan	ACN
Prevent organ rejection in transplant	Sapropterin	Water
Glucose deficiency	Tacrolimus	ACN, water
Restructure proteins	Teneligliptin	ACN, water
Asthma	Urea adducts	Water
	Vilanterol	ACN, water